

Decent Coaching Classes
VIII CBSE
Mind Your Math - VI

Subject : Mathematics (Algebraic Expressions & Polynomials, Factorisation)

Marks : 30

Note : All questions are compulsory. Section C has internal choice in each question.

Section A — Multiple Choice Questions (1 mark each)

1. The coefficient of x^2 in the expression $7x^2 - 5x + 3$ is : [1]
(a) 3 (b) -5 (c) 7 (d) 1
2. Which of the following is a trinomial? [1]
(a) $5x^2y$ (b) $x^2 + y^2$ (c) $a - b + c$ (d) $3x$
3. The product of $3x$ and $(2x - 5)$ is : [1]
(a) $6x - 15$ (b) $6x^2 - 15x$ (c) $6x^2 - 5x$ (d) $5x - 15$
4. On subtracting $(3a - 2b)$ from $(5a + b)$, we get : [1]
(a) $2a - 3b$ (b) $2a + 3b$ (c) $8a - b$ (d) $-2a + 3b$
5. The common factor of $12x^2y$ and $18xy^2$ is : [1]
(a) $6xy$ (b) $6x^2y^2$ (c) $12x$ (d) $18y$
6. $x^2 + 6x + 9$ is a perfect square of : [1]
(a) $(x + 3)$ (b) $(x - 3)$ (c) $(x + 9)$ (d) $(x + 6)$

Section B — Very Short Answer Questions (2 marks each)

7. Write the coefficient of a^2 and the constant term in the expression : $3 + a^2 - 5b^2$ [2]
8. Find the product : $(3a + 5b)(5a + 7b)$ [2]
9. Factorise by taking out the common factor : $15x^2y - 20xy^2$ [2]
10. Factorise : $x^2 + 10x + 25$ [2]

Section C — Short Answer Questions (4 marks each, with internal choice)

11. Find the product : $(a^2 - 3a + 7) \times (2a + 3)$ [4]

OR

Find the product : $(a^2 - 5a + 8) \times (a^2 + 2)$

12. Simplify : $(2x + 7y)(x - 3y) - (9x + 4)(3x - 2)$ [4]

OR

Simplify : $(5x^2 - y^2)(x^2 + 3y^2) + (2xy)^2$

13. Factorise : $4x^2 + 20x + 25$ [4]

OR

Factorise : $9a^2 - 12ab + 4b^2$

Section D — Case Study Based Question (4 marks)

14. The area of a rectangular garden plot is given by the expression $6x^2y + 9xy^2$ (in sq. m). Based on this, answer the following :

- (a) Factorise the given expression by taking out the common factor. [2]
- (b) Hence, find the numerical value of the area when $x = 2$ and $y = 1$. [1]

(c) Verify that the original expression and your factorised expression give the same value for $x = 2$, $y = 1$. [1]